

1 Downloading Operation Data

Overview

Message type: LOIPLO
 IDoc type: LOIPLO01
 Segments: E2PLAFL001
 E2PLOPL001
 E2KBEDL
 E2RESBL

Segment E2PLAFL001

Segment E2PLAFL001 includes data of the planned order header.

Field name	T	L	D	Description	Usage in HYDRA
ABMNG	QUAN	14	3	Reduced quantity in planned order	Not used
AUFFX ANR.FU:29	CHAR	1		Fixing flag in planned order data	Not used
AUFNR	CHAR	12		Serial order number	Not used
AVMNG ANR.SGR:AUSB	QUAN	14		Planned scrap quantity	OH.Target scrap in base quantity unit
BESKZ	CHAR	1		Procurement type	Not used
DISPO ANR.DISP	CHAR	3		Material controller	OH.MRP controller
EKORG	CHAR	4		Purchasing organization	Not used
GSBTR	DATS	8		Total confirmation date of planned order after ATP check of components	Not used
GLTRS	DATS	8		Scheduled end time	Not used
GLUZS	TIMS	6		Earliest scheduled end time: Execution	Not used
GSMNG ANR.SGR:GUTB	QUAN	14		Planned order quantity	OH.Target quantity in base quantity unit
GSTRS	DATS	8		Scheduled start time	Not used
GSUZS	TIMS	6		Earliest scheduled start time: Execution (time)	Not used
KAPFX	CHAR	1		Flag: Planned order - capacities planned	Not used
KDAUF ANR.KDAUF	CHAR	10		Sales order number	OH.Sales order
KDPOS ANR.KDPOS	CHAR	6		Item number in sales order	OH.Sales order item
LGORT ANR.LGORT	CHAR	4		Storage location	To be used as receiving storage location in confirmation
MATNR ANR.ATK	CHAR	18		Material number	OH.Article

Field name	T	L	D	Description	Usage in HYDRA
MDACC	CHAR	4		Planned order handling action	Not used
MDACD	DATS	8		Planned order action date	Not used
MDACH	CHAR	2		Planned order handling action control	Not used
MDACT	TIMS	6		Planned order handling action time	Not used
MDPBV	CHAR	1		Planned order confirmation availability	Not used
MEINS ANR.SGE:GUTB	UNIT	3		Base quantity unit	OH.Base quantity unit OP.Base quantity unit
PAART ANR.FU:45	CHAR	4		Order type	Not used
PALTR	DATS	8		Resolution date (explosion date)	Not used
PEDTR ANR.DATSE	DATS	8		Order finish date in planned order	OH.Order finish date
PERTR	DATS	8		Planned creation date in planned order	Not used
PLNAL	CHAR	2		Group counter	OH.Work plan
PLNNR ANR.APNR	CHAR	8		Key of planned group	OH.Work plan
PLNTY	CHAR	1		Planned type	Not used
PLNUM ANR.SAPAUNR	CHAR	10		Planned order number	HYDRA order number according to configuration (*1)
PLSCN	NUMC	3		Long-term planning scenario	Not used
PLWRK	CHAR	4		Planning plant	Not used
PSPEL ANR:PRJNR	NUMC	24		PSP element	OH.Project number
PSTTR ANR.DATFB	DATS	8		Order start date in planned order	OH.Order start date
PWWRK ANR.WERK:S	CHAR	4		Production plant in planned order	Used as actual plant in confirmation
SEQNR	NUMC	14		Order sequence number	Not used
SERNR	CHAR	8		Serial number	Not used
SOBES	CHAR	1		Special procurement type	Not used
SOBKZ	CHAR	1		Special inventory flag	Not used
STLFX	CHAR	1		Fixing flag for BOM explosion	Not used
UMSKZ	CHAR	1		Implementation flag of planned order	Not used
VERID ANR.FERTVER	CHAR	4		Production version	Stored for confirmation
VFMNG	QUAN	14		Planned order of confirmed quantity	Not used
WEBAZ	DEC	5		Processing time for goods receipt in days	Not used
MATNR_EXTERNAL	CHAR	40		Long material number	Not used
MATNR_VERSION	CHAR	10		Version number for MATNR field	Not used
MATNR_GUID	CHAR	32		External GUID for MATNR field	Not used

Segment E2PLOPL001

The segment E2PLOPL001 includes operation data.

Field name	T	L	D	Description	Usage in HYDRA
VORN ANR.SAPVGNR	CHAR	4		Operation number	HYDRA operation number according to configuration (*1)
PVZNR	CHAR	4		Superordinate operation	Not used
ARBEH	UNIT	3		Work unit	Not used
ARBEI	QUAN	8		Operation work	Not used
ARBID ANR.MNR	NUMC	8		Workplace ID	Target workplace OP.Machine and/or OP.Machine group (see note)
DAUNE	UNIT	3		Unit duration normal	Not used
DAUNO	QUAN	6		Normal operation duration	Not used
FSEDD	DATS	8		Earliest scheduled end date: Execution (date)	Not used
FSEDZ	TIMS	6		Earliest scheduled end time: Execution (time)	Not used
FSAVD	DATS	8		Earliest scheduled start date: Execution (date)	Not used
FSAVZ	TIMS	6		Earliest scheduled start time: Execution (time)	Not used
FSSAD	DATS	8		Earliest scheduled start date: tear down/retool (date)	Not used
FSSAZ	TIMS	6		Earliest scheduled start time: tear down/retool (time)	Not used
FSSBD	DATS	8		Earliest scheduled start date: Processing (date)	Not used
FSSBZ	TIMS	6		Earliest scheduled start time: Processing (time)	Not used
PLNFL	CHAR	6		Sequence	Not used
SSEDD	DATS	8		Latest scheduled end date: Execution (date)	Not used
SSEDZ	TIMS	6		Latest scheduled end time: Execution (time)	Not used
SSAVD	DATS	8		Latest scheduled start date: Execution (date)	Not used
SSVAZ	TIMS	6		Latest scheduled start time: Execution (time)	Not used
SSSAD	DATS	8		Latest scheduled start date: tear down/retool (date)	Not used
SSSAZ	TIMS	6		Latest scheduled start time: tear down/retool (time)	Not used
SSSBD	DATS	8		Latest scheduled start date: Processing (date)	Not used
SSSBZ	TIMS	6		Latest scheduled start time: Processing (time)	Not used
USR04 ANR.TLG	QUAN	15		User field 04	Partitioning (see note)
USR05 ANR.SZY	QUAN	15		User field 05	Target cycle (see note)
USE04	UNIT	3		Unit user field 04	Not used
USE05	UNIT	3		Unit user field 04	Not used
VGE01	UNIT	3		Unit of activity 1	Not used
VGE02	UNIT	3		Unit of activity 2	Not used

Field name	T	L	D	Description	Usage in HYDRA
VGE03	UNIT	3		Unit of activity 3	Not used
VGE04	UNIT	3		Unit of activity 4	Not used
VGE05	UNIT	3		Unit of activity 5	Not used
VGE06	UNIT	3		Unit of activity 6	Not used
VGW01 ANR.VGW01	QUAN	10		Default value activity 1	Not used
VGW02 ANR.VGW02	QUAN	10		Default value activity 2	Not used
VGW03 ANR.VGW03	QUAN	10		Default value activity 3	Not used
VGW04 ANR.VGW04	QUAN	10		Default value activity 4	Not used
VGW05 ANR.VGW05	QUAN	10		Default value activity 5	Not used
VGW06 ANR.VGW06	QUAN	10		Default value activity 6	Not used
XDISP	CHAR	1		Flag: Operation/KBED is planned	Not used
ANZMA ANR.MBVERH:NORM	DEC	7	2	Number of employees	Not used

Segment E2KBEDL

Field name	T	L	D	Description	Usage in HYDRA
BEDID	NUMC	12		Serial number of capacity requirements record	Not used
BEDZL	NUMC	8		Internal counter	Not used
CANUM	NUMC	4		Counter of capacity requirements record	Not used
BEDKZ	CHAR	1		Availability indicator	Not used
KABRREST	FTP	22		Remaining quantity clearing/retooling capacity demand	Not used
KABRSOLL	FTP	22		Clearing/retooling capacity demand	Not used
KAPAR	CHAR	3		Capacity category	Not used
KAPID	NUMC	8		Capacity (number)	Only records of capacity type "001" are transferred.
KBEAREST	FTP	22		Remaining quantity processing capacity requirements	Not used
KBEASOLL	FTP	22		Processing capacity requirements	Processing duration (target for RPA11)
KEINH	UNIT	3		Unit of capacity requirements	Unit for processing time and setup time (*2)
KRUEREST	FTP	22		Remaining quantity setup capacity requirements	Not used
KRUESOLL	FTP	22		Setup capacity requirements	Setup duration (target for RPA7) (*2)

OH.xxx Field xxx in HYDRA order header

OP.xxx Field xxx in HYDRA operation

(*1) For more information on generating the HYDRA order number, please see below.

(*2) The contents of these fields must not exceed a duration of 590 hours.

Notes on the operation structure

HYDRA order number (*1)

The HYDRA order number consists of a configurable part of the SAP key fields **E2PLAFL.PLNUM**, **E2PLOPL.VORNR**. The following steps are carried out to make sure the order number is transferred to HYDRA in a standardized manner.

- When the CHAR fields are transferred, the system converts blank characters to "0", since the barcode is also assigned "0" instead of blank characters.
- Two leading places are added to the SAP order number in order to calculate the HYDRA order number. Consequently, HYDRA uses a number that is extended by two places compared to the original SAP planned order number. This is taken into account if quantities are uploaded to SAP for the planned order (series production).
- By default, the following conventions apply:
→ PLNUM (=order) C8, VORNR (=operation) C4.

Operation quantity

The SAP IDoc type LOIPLO01 does not provide an explicit operation quantity. SAP only provides a target yield and a target scrap quantity in the order header. This quantity is transferred to each operation that is sent with this order. Any scrap quantities of the preceding operations cannot be integrated.

Overview - how the user fields are used:

By default, HYDRA processes the user-specific fields USR04, USR05, USE04 and USE05 as follows:

User field 04 (USR04):

HYDRA partitioning/cavity

User field 05 (USR05):

HYDRA target cycle in seconds per 1000 machine strokes

User field 04 unit (USE04):

Not used

User field 05 unit (USE05):

Not used

HYDRA partitioning

If the user field USR04 includes a numeric value, the system enters this value as the partitioning in the operation.

If this is not the case or if the value is "0", the system automatically enters the value 1 as the partitioning.

BDE cycle time

The system attempts to take the target cycle from the field E2PLOPL001.USR05. If this is not possible as no target cycle is entered (i.e. the field is "0"), the system uses the following formula to calculate the target cycle using the processing time value of the segment E2KBEDL.KBEASOLL. To do so, this segment must include a capacity type record "001":
$$\text{int}((\text{E2KBEDL.KBEASOLL} / (\text{E2PLAFL.GSMNG} / \text{E2PLOPL001.USR04})) * 1000)$$

Standard times

In HYDRA the content of the field E2KBEDL.KBEASOLL specifies the standard time for machine assignment (target for RPA 11).

In HYDRA the content of the field "E2KBEDL.KRUESOLL" specifies the standard time for the machine setup (target for RPA 7).

Information on the workplace - ARBID field:

- The workplace ID (ARBID) is not the workplace number as it is stored in the work plan but the unique ID of the workplace. Data is collected in HYDRA and uploaded to SAP R/3 PP-REM for the workplace entered in this field.